Radiative cancellations and localized linear modes in three-dimensional sextic Q-balls

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Abstract

We study localized linear excitations of a complex scalar Q-ball in three spatial dimensions with potential $\mathcal{U}(S) = S - S^2 + S^3/2$. In a regime with one propagating and one evanescent exterior channel, we assemble computer-assisted existence arguments for two radial modes and one dipole mode on the infinite radial domain. Analytic exterior estimates and ball-arithmetic interior enclosures distinguish these results from finite-box linewidth minima. A numerical radial sequence, with candidates reported through index eighteen, motivates a thin-wall phase-matching description. An independently computed planar transmission zero yields an unfitted spacing candidate; identifying this value with a radial limiting spacing remains open. A second-harmonic analysis separates one dipole monopole outgoing coefficient from zero under explicitly audited exact-mode and enclosure hypotheses. It obstructs a specified uniformly localized smooth continuation for every linked dipole polarization with $\boldsymbol{p}^T\boldsymbol{p} \neq 0$; circular polarization requires a separate quadrupole certificate. Linear existence, numerical spacing and nonlinear continuation are treated as distinct statements.

1 Question and scope

A small resonance width, an exactly localized linear mode and a localized finite-amplitude oscillation require different evidence. We examine this separation for a complex scalar Q-ball, reporting three existence enclosures, a broader numerical sequence and a wall-scattering interpretation of its spacing. The sequence label n is a continuation index, not a certified node count or a completeness statement. A bound state in the continuum (BIC) denotes a localized linear mode in an available propagating-channel regime [26]. Uncertified locations will be called candidates. All quantities are dimensionless.

1.1 Relation to Q-ball excitations and continuum trapping

Q-balls are localized scalar configurations carrying a conserved global charge [1]. Battye and Sutcliffe study collisions, charge transfer and fission in one, two and three spatial dimensions in the same sextic model, after unit conversion [13]. Kovtun, Nugaev, and Shkerin formulate the coupled sideband problem $\omega \pm \rho$ and analyze three-dimensional vibrational modes [2]. Azatov, Ho, and Khalil treat scattering in the sextic family, as well as the complex-plus-real-field Friedberg–Lee–Sirlin model [7, 29]. The coupled radial operator is established prior work; the contribution here concerns particular localized solutions of it, with regular origin conditions and controlled infinite tails.

Chen, Andersson, and Li analyze three-dimensional Q-ball perturbations for the rescaled potential $V(s) = s^2 - s^3 + \kappa s^4$, where s denotes the field modulus [31]. Their coupled-sideband classification provides context, but concerns a different potential. Our claim is the certification of three particular localized linear modes of the sextic model, including cancellation of both oscillatory open-channel components. Among the works reviewed here, we did not identify a predecessor for these specific modes. This limited statement establishes neither global priority nor a contradiction with the spectrum studied by Chen and collaborators.

Zhang, Zhou, and Zhu obtain analytic scattering solutions from stepwise Q-ball profiles [8]. Their interior dispersion and thin-wall phase structure are directly relevant to Eq. (16), although their propagating scattering regime has two exterior open channels. Ciurla et al. distinguish bound, half-propagating and quasinormal modes in $1+1$ dimensions [3]. Their half-propagating interval $1-\omega<\rho<1+\omega$ has the same channel thresholds as ours, but their long-lived example retains a small radiative tail. Evslin et al. analyze paired Floquet modes of small-amplitude, one-dimensional Q-balls [10]; coupling of a discrete component to its propagating partner produces Feshbach-type quasinormal modes. These one-dimensional calculations do not transfer the spherical origin condition or radial measure to three dimensions.

Numerically assisted spectral exclusion arguments for NLS solitary waves include those of Marzuola and Simpson and Asad and Simpson [24, 25]. Cuccagna and Maeda impose absence of embedded eigenvalues as hypothesis H4 and describe the broader ground-state expectation as unproved in their 2020 survey [32, Remark 3.2]. Li and Yang subsequently prove absence in the essential spectrum for the full linearized operator about the ground state of the three-dimensional focusing cubic NLS [33, Theorem 1.1]. This is a different equation and operator from our relativistic Q-ball frequency pencil. A nodeless Q-ball profile alone does not establish the relevant variational ground-state class. The present model-specific existence claim therefore does not contradict that NLS theorem or refute the survey's expectation in its stated setting.

A bound state in the continuum (BIC) is a spatially localized linear state at a frequency also admitting extended radiation modes [26]. Interference-induced trapping is an established mechanism: Friedrich and Wintgen relate interacting resonances to vanishing decay width [4], and Yu and Lu give an existence theorem for a three-channel Schrödinger problem with one open channel [5]. Its small-coupling assumptions and finite-range potential differ from the present two-channel pencil with an infinite profile tail. Related resonant reflection occurs for discrete breathers, optical solitons and condensate domain walls [19, 20, 21].

Winding numbers for BICs are also established: Zhen et al. use the winding of a real far-field polarization vector in the two-dimensional Bloch wave-vector plane of a photonic slab [35]. Their symmetry assumptions support a charge-conservation and robustness statement. Our winding of a matching map over Q-ball parameters is a numerical zero-finding diagnostic; it does not inherit that photonic robustness theorem.

The planar transmission zero and interior phase matching admit a *Fabry–Pérot interpretation.* Hsu et al. describe BICs formed by perfect resonant reflection together with a round-trip phase condition, including a resonator paired with a reflecting boundary [26]. In the thin-wall picture used here, the reflecting wall and the regular centre supply the analogous radial cavity. This is an interpretation of the wall and phase calculations, rather than an additional existence result for the curved Q-ball. In particular, the planar zero alone does not certify a spherical mode. Sequences of radiation cancellations and WKB matching are also known for nonlinear embedded solitons [11]; their model and scaling parameter differ from those of the present sequence of linear modes on changing backgrounds.

Nonlinear radiation obstructions have a separate history. Segur and Kruskal find beyond-all-orders radiation excluding small-amplitude true breathers in the one-dimensional ϕ^4 model [27]. Sigal studies instability of periodic and quasiperiodic wave solutions under generic nonlinear perturbations [28]; Soffer and Weinstein prove radiation damping under a Fermi-golden-rule hypothesis [12]. Manton and Merabet derive second-harmonic radiation with amplitude of order A^2 and power of order A^4 from a kink's internal mode in $1+1$-dimensional ϕ^4 theory [36, Appendix A]. The harmonic mechanism and power counting are known; their coefficients and perturbative decay law are not transferred to our three-dimensional Q-ball. The present contribution in this direction is the explicit error budget separating a particular second-harmonic coefficient from zero. Its continuation obstruction has the stated hypotheses and is not a dynamical decay theorem. Oscillon decay minima [9] and the second-harmonic radiation zeros of a critical-bubble internal mode [23] further illustrate why suppression at one harmonic does not establish an exactly localized nonlinear oscillation.

Lührmann et al. combine analytic estimates and interval arithmetic to certify vortex profiles, spectral data and nonzero Fermi-golden-rule coefficients in the self-dual abelian Yang–Mills–Higgs model in two spatial dimensions [34, Theorem 1.1, Remark 1.2 and Section 2]. Their internal mode is a gap eigenvalue. This is a methodological precedent for rigorous spectral and radiation-coupling computations. Here the distinct target is a particular outgoing second-harmonic coefficient associated with a localized linear Q-ball mode in three dimensions. Its conditional continuation obstruction remains separate from a nonlinear decay, lifetime or background-stability theorem.

Background stability is likewise distinct from mode localization. Classical stability results for charged scalar solitons and the Vakhitov–Kolokolov slope criterion provide the appropriate context [29, 30]. Applying such a criterion requires its model-dependent hypotheses and charge convention; the certificates here do not determine $dQ/d\omega$ at the three backgrounds. The literature comparisons delimit the claimed contribution without establishing exhaustive priority or an infinite ladder.

2 Field model and two-channel linearization

With metric signature $(+,-,-,-)$, define

$$\mathcal{L} = \partial_\mu \phi^* \partial^\mu \phi - \mathcal{U}(|\phi|^2), \qquad \mathcal{U}(S) = S - S^2 + \beta S^3, \qquad \beta = \tfrac{1}{2}. \tag{1}$$

The free scalar mass is one. A spherical stationary background is $\phi_0 = e^{\mathrm{i}\omega t} f(r)$ with real f, satisfying

$$f'' + \frac{2}{r} f' = (1-\omega^2) f - 2f^3 + 3\beta f^5, \quad f'(0) = 0, \quad f(r) \longrightarrow 0. \tag{2}$$

For definiteness the charge sign is chosen so that $Q = 2\omega \int f^2 \mathrm{d}^3x > 0$ when $\omega > 0$. The thin-wall frequency threshold is $\omega_c^2 = \min_{S>0} \mathcal{U}(S)/S = 1 - 1/(4\beta) = 1/2$.

Use a real orthonormal spherical-harmonic basis and write

$$\delta\phi = e^{\mathrm{i}\omega t}[a(r) e^{\mathrm{i}\rho t} + b(r) e^{-\mathrm{i}\rho t}] Y_{lm}(\hat{x}), \qquad A = ra, \quad B = rb, \tag{3}$$

for real-frequency, real-radial-function modes. In a complex harmonic basis the second sideband instead carries Y_{lm}^*. Let $S = f^2$, $D = 1 - 4S + 9\beta S^2$, and $C = -2S + 6\beta S^2$. The radial equations are

$$A'' = \left[D + \frac{l(l+1)}{r^2} - (\omega+\rho)^2 \right] A + CB, \tag{4}$$

$$B'' = CA + \left[D + \frac{l(l+1)}{r^2} - (\omega-\rho)^2 \right] B. \tag{5}$$

Regular solutions satisfy $A, B = O(r^{l+1})$ at the origin. In the range $1-\omega < \rho < 1+\omega$, define

$$k^2 = (\omega+\rho)^2 - 1 > 0, \qquad \kappa_c^2 = 1 - (\omega-\rho)^2 > 0, \qquad \kappa_0^2 = 1 - \omega^2 > 0. \tag{6}$$

The A channel is open and the B channel closed. A localized mode requires cancellation of both oscillatory exterior components of A and exclusion of the growing component of B. The asymptotic continuous-frequency branches start at $|\rho| = 1-\omega$, with the second channel opening at $|\rho| = 1+\omega$; these are the standard Q-ball sideband thresholds [2, 3]. Here "embedded" specifies an exactly localized solution in this continuous-frequency range. The certificates establish a partial-wave statement, rather than a separate theorem about the full evolution generator's essential spectrum or algebraic multiplicity.

The spatial dimension is three; the two sidebands are field components, not additional spatial dimensions. Spherical reduction retains the three-dimensional measure and centrifugal term. It is neither a one-dimensional field theory nor a test of all three-dimensional perturbations. In one spatial dimension there is no spherical centrifugal term; in two dimensions both the radial measure

and angular decomposition change. The planar wall below is a local thin-wall approximation to the three-dimensional profile. The separate two-dimensional numerical study is retained in the Supplement; profiles and charge normalizations are not transferred between dimensions. Higher spatial dimensions are outside the present evidence.

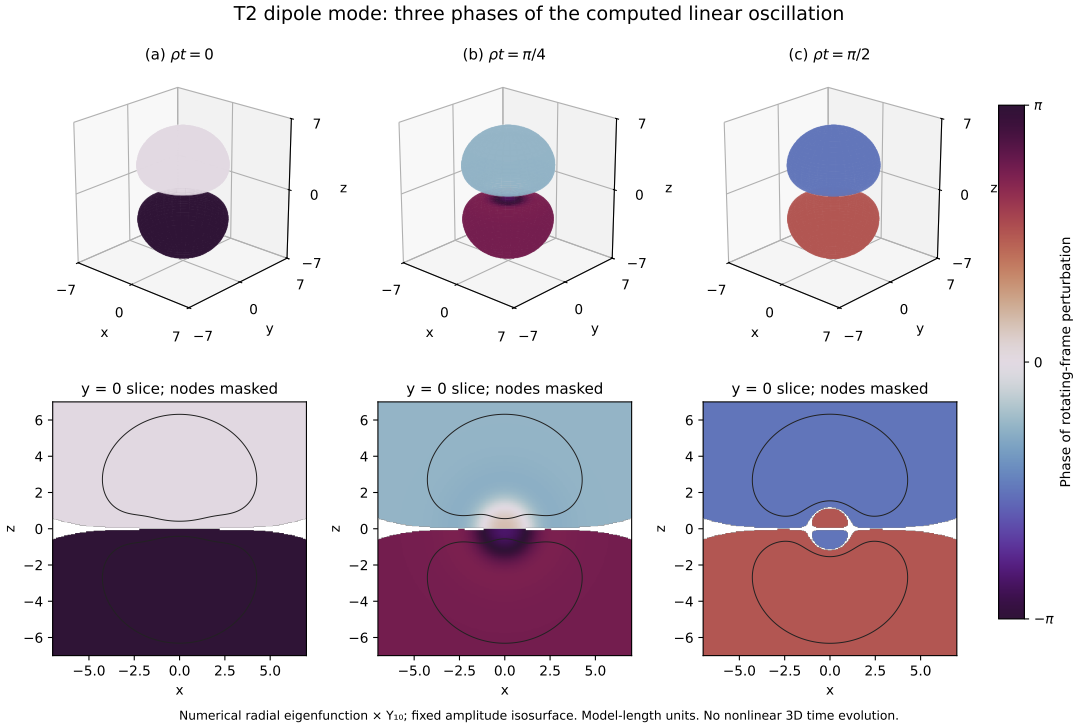


Figure 1: Numerical reconstruction near the enclosed dipole mode C2, with angular factor $Y_{10} = \sqrt{3/(4\pi)}z/r$. Top: three phases of the linear perturbation; bottom: $y = 0$ sections with phase masked below 1.5% of the common displayed maximum. The time factors are the two linear sidebands, not a nonlinear evolution. The stored visualization's historical label is mapped to C2 in the archive figure inventory.

3 Computer-assisted existence results

We distinguish the three computer-assisted existence results below from the numerically observed ladder. The proof establishes a localized mode at a parameter pair inside each specified box. It does not certify every observed radiation minimum, an infinite ladder, or a nonlinear nonradiating solution.

3.1 Model and precise statement

Consider the complex Klein–Gordon equation in three spatial dimensions,

$$\partial_t^2\phi - \Delta\phi + U'(|\phi|^2)\phi = 0, \qquad U(S) = S - S^2 + \tfrac{1}{2}S^3. \tag{7}$$

A stationary Q-ball has the form $\phi_0 = e^{i\omega t}f(r)$, where

$$f'' + \frac{2}{r}f' = (1-\omega^2)f - 2f^3 + \tfrac{3}{2}f^5, \qquad f(0) = a, \quad f'(0) = 0. \tag{8}$$

We use a real orthonormal basis $Y_{\ell m}$ of spherical harmonics and write a linear perturbation as

$$\delta\phi = e^{i\omega t}\left[\frac{A(r)}{r}e^{i\rho t} + \frac{B(r)}{r}e^{-i\rho t}\right]Y_{\ell m}. \tag{9}$$

In a complex harmonic basis the second sideband instead contains $Y^*_{\ell m}$. With $S = f^2$, substitution in the linearization of (7) gives

$$\begin{aligned} A'' &= \left[1 - 4S + \tfrac{9}{2}S^2 - (\omega+\rho)^2 + \frac{\ell(\ell+1)}{r^2}\right] A + CB, \\ B'' &= \left[1 - 4S + \tfrac{9}{2}S^2 - (\omega-\rho)^2 + \frac{\ell(\ell+1)}{r^2}\right] B + CA, \qquad C = -2S + 3S^2. \end{aligned} \tag{10}$$

Regularity requires $A, B = O(r^{\ell+1})$. Define

$$\kappa_0 = \sqrt{1-\omega^2}, \qquad k = \sqrt{(\omega+\rho)^2 - 1}, \qquad \kappa_c = \sqrt{1-(\omega-\rho)^2}. \tag{11}$$

Computer-assisted existence statement. For each row of Table 1, the corresponding certificate encloses a quadruple $(a_*, c_*, \rho_*, \omega_*)$ with $c_* > 0$ such that (8) has a positive global solution satisfying $re^{\kappa_0 r} f(r) \to c_*$, and (10) has a nonzero real solution regular at the origin with

$$|A(r)| + |B(r)| \le K_J e^{-\kappa_c r} \quad (r \ge L), \qquad e^{\kappa_c r} B(r) \longrightarrow 1. \tag{12}$$

The resulting spatial fields in (9) are smooth and exponentially localized. The certification is conditional on the correctness of the stated interval-arithmetic implementation and its underlying arithmetic libraries.

Case	ℓ	Enclosure of ω_*^2	Enclosure of ρ_*
C0	0	$0.797676787111865191750 \pm 4.8\,10^{-22}$	$1.744617544837398735781 \pm 6.8\,10^{-22}$
C1	0	$0.6851289044582160933 \pm 2.5\,10^{-20}$	$1.6903565973281434568 \pm 4.7\,10^{-20}$
C2	1	$0.7544960137826372331 \pm 7.8\,10^{-20}$	$1.826342030181069956 \pm 3.0\,10^{-19}$

Table 1: The three certified parameter boxes, displayed as outward-rounded decimal enclosures. The proof uses the full exact dyadic boxes in (a, c, ρ, ω), not these two-coordinate summaries. Their numerical ladder assignments are $(\ell, n) = (0,1), (0,2), (1,1)$; the labels n are not separately certified nodal counts.

For the enclosed parameter pair and its fixed radial angular channel, every solution of (10) that is square integrable on $[L, \infty)$ is a scalar multiple of the constructed solution. This is *geometric radial simplicity*. It neither establishes algebraic simplicity of a dynamical generator nor excludes Jordan chains. For C2, the three-dimensional $\ell = 1$ angular multiplet remains degenerate. The translation mode at $\rho = 0$ does not belong to the fixed nonzero frequency considered here.

The term *embedded* is used operationally: $k^2 > 0$ and $\kappa_c^2 > 0$, so a free asymptotic channel is open, while the constructed mode is exponentially localized and has no oscillatory radiation tail. We do not identify the essential spectrum of a fully specified operator pencil in this proof. The perturbation is periodic after removing $e^{i\omega t}$; the two laboratory frequencies are $\omega \pm \rho$, with no claim about their rational commensurability.

3.2 Infinite-domain construction and inclusion

The matching radius L joins an interval-integrated core to an analytic infinite exterior. The potential and its coupling are retained beyond L. Appendix A gives the complete analytic lemmas used below, including the origin start and the error bounds for every Taylor step.

For the profile, $\widetilde{u} = re^{\kappa_0 r} f$ satisfies

$$\widetilde{u}(r) = c + \int_r^\infty \frac{1 - e^{-2\kappa_0(s-r)}}{2\kappa_0} \left(-2f(s)^2 + \tfrac{3}{2}f(s)^4\right) \widetilde{u}(s)\, \mathrm{d}s. \tag{13}$$

The certified exterior constants give a uniform contraction in a positive ball about c. For the mode, the free open and closed Green kernels produce a Volterra equation with integrable coefficients. Ordered iteration has factorial error majorants, and yields a normalized exponentially decaying Jost solution. The $\ell = 1$ construction retains the exact Riccati functions and the centrifugal term. These constructions supply continuous corrections to both the profile Cauchy data and the mode Cauchy data at L, uniformly on the parameter box.

3.3 Finite-dimensional certificate

Let $z = (a, c, \rho, \omega)$. The four-component matching map is $H = H_0 + E$. Its first two components match profile value and derivative; the last two are scaled Wronski forms $W(\Psi, R_i) = \Psi \cdot R_i' - \Psi' \cdot R_i$, where R_1, R_2 are the two regular origin solutions. The nominal map H_0 uses the free Jost data and E bounds the infinite-tail correction. For $\ell = 0$, these pairings equal the origin values of Ψ. For $\ell = 1$, the regular behaviors $r^2 e_i$ pair with singular behaviors $r^{-1} e_i/3$ to give δ_{ij}. Thus their joint vanishing is precisely regularity of the Jost solution at the origin.

The core integration encloses $H_0(z_0)$ and $DH_0(Z)$ using complex analytic a priori bounds and Taylor remainders. At the origin, the profile integral equation has weight mass $r^2/6$; for $A = r^2\alpha$, $B = r^2\beta$ the $\ell = 1$ weight mass is $r^2/10$. Parameter derivatives are integrated as part of the same enclosed system. All step choices and preconditioners become exact dyadic input; acceptance uses directed arithmetic.

Write $Z = z_0 + [-\delta, \delta]$, let D enclose $DH_0(Z)$, and let $|E_i| \le \epsilon_i$ throughout Z. A fixed dyadic matrix Y gives

$$\begin{aligned} z_0 - Y H_0(z_0) + (I - YD)[-\delta, \delta] + |Y|[-\epsilon, \epsilon] &\subset \operatorname{int} Z, \\ \max_i \sum_j |I - YD|_{ij} \frac{\delta_j}{\delta_i} &< 1. \end{aligned} \tag{14}$$

The first inclusion makes $z \mapsto z - YH(z)$ a continuous self-map of Z. Brouwer gives a fixed point; the weighted matrix inequality makes Y invertible, so $H(z_*) = 0$. This proof requires continuity, but no derivative bound, for E and does not assert uniqueness of z_*. Positive core enclosures, a minimum-principle argument on the remaining finite interval, and the positive exterior fixed point establish $f > 0$. Finally $2\kappa_0 > \kappa_c$ permits two bounded oscillatory comparison solutions at infinity. Their nonzero mutual Wronskian and the four-dimensional symplectic Cauchy-data space imply that every square-integrable solution is a multiple of the decaying Jost solution. Appendix A.6 supplies this simplicity proof explicitly.

3.4 Certificate contents and verification scope

The defining certificates use $L = 40$ for C0, $L = 32$ for C1, and $L = 36$ for C2, with 256-bit ball arithmetic. The preserved historical records identify these three JSON certificates by case, filename and SHA-256 checksum. A distribution copy with redacted execution metadata is a derivative: the archive manifest lists its checksum separately from the historical checksum, while retaining the numerical fields. The full dyadic boxes, channel bounds, tail constants, inclusion results and integration-step records are the proof inputs. The fresh primary-certificate reproduction uses the preserved historical proof-code bytes. The distribution provides separately hashed code derivatives with redacted comments and one metadata string in the unused starting-value-search mode; `CODE-HASH-MAP.tsv` records their correspondence. These changes leave the certificate execution path unchanged. A filename recorded by an old driver is resolved through its recorded hash, rather than assumed to identify a later file with the same name.

Additional runs use larger matching radii and 320-bit arithmetic. C2's original repeat at $L = 40$ failed the inclusion test; a subsequent run with an enlarged c-radius passed after reevaluating all bounds on that box. This is an adaptive additional certificate. The C2 existence claim rests on the first successful certificate alone.

For C0, $2\kappa_0 > \kappa_c$ was recorded and separately checked but was absent from the historical aggregate success flag. This documented flag omission is retained. In this revision a separate ball-arithmetic check reconstructs the exact dyadic parameter box from the reproduced C0 receipt and verifies $2\kappa_0 - \kappa_c > 0$ throughout that box, together with $\rho > \omega$. It binds the checked receipt and the checker by their checksums; neither the historical driver nor its success flag is amended. The C1/C2 drivers include the inequality explicitly. Repeated runs and code review use the same implementation and do not supply an independent second implementation. Correctness of

Python–FLINT, Arb/Acb and the recorded execution remains part of the computational trust base.

The arithmetic uses midpoint–radius enclosures [14, 15, 16]; the finite-dimensional inclusion is of Krawczyk type [17], with the explicit Brouwer argument above for the continuous tail correction. The three linear existence results leave background stability, algebraic spectral simplicity and certification of the full empirical ladder open.

4 Numerical radial sequence

A width minimum is a numerical candidate; a resolved winding of a specified two-component matching map adds numerical evidence; an interval enclosure with analytic tails establishes infinite-domain existence. Table 2 keeps these levels separate. The winding concept has a photonic BIC precedent [35]; here it is applied to a matching map in parameter space, without transferring the photonic symmetry-based robustness theorem. The refined matching calculation for indices 6–10 used radial steps 0.04 and 0.02. Its locations interpolate parameter samples separated by 2.5×10^{-4}. For indices 7–10, the small inter-grid difference (2.3×10^{-8} or less) is not a validated location error. The reported interpolation uncertainty is a few 10^{-6}. For indices 11–14, the archived linewidth reconstruction used signs chosen to match neighboring slopes. Those rows remain weaker evidence; no additional matching calculation was made for this revision.

Table 2: Radial candidates at $\beta = 1/2$. Displayed numerical digits are not interval enclosures. Certificates C0/C1 are given in Section 3; the remaining entries are finite-domain numerical evidence. The last three error estimates are empirical.

n	ω^2	Evidence
1	$0.797676787\ldots$	Existence enclosure C0
2	$0.685128904\ldots$	Existence enclosure C1
3	~ 0.631450	Resolved numerical winding
4	~ 0.601421	Resolved numerical winding
5	~ 0.582417	Sign change; winding unresolved
6	~ 0.569360	Resolved numerical winding
7	~ 0.559848	Resolved numerical winding
8	~ 0.552619	Resolved numerical winding
9	~ 0.546940	Resolved numerical winding
10	~ 0.542364	Resolved numerical winding
11	~ 0.53860	Linewidth reconstruction
12	~ 0.535449	Linewidth reconstruction
13	~ 0.532772	Linewidth reconstruction
14	~ 0.530470	Linewidth reconstruction
15	~ 0.52846934	Subsequent resolved winding
16	$0.52671542 \pm 2 \times 10^{-7}$	Resolved winding, final bracket rule failed
17	$0.52516277 \pm 1.3 \times 10^{-6}$	Resolved winding, final bracket rule failed
18	$0.5237875 \pm 4 \times 10^{-6}$	Crossing count +1; phase unresolved

The subsequent calculation reproduced index 15 near 0.5284693357. For 16 and 17, resolved numerical windings are +1 and −1. For 18, the crossing count and sampled phase winding both give +1, but the maximum phase jump 0.585 radians fails the resolution rule. All three miss the prescribed final-bracket acceptance criterion; the empirical errors include roundoff sensitivity. This unsuccessful combined acceptance rule is retained, even where individual diagnostics are informative. There is no certificate or completeness result for these additional candidates.

Prediction history. A sequential extrapolation was updated after each group of targets. A separate four-parameter thin-wall fit used indices $5, 7, 8, 9$; its author reported separation from later target values, which were already known to the coordination team. The frozen file of 30

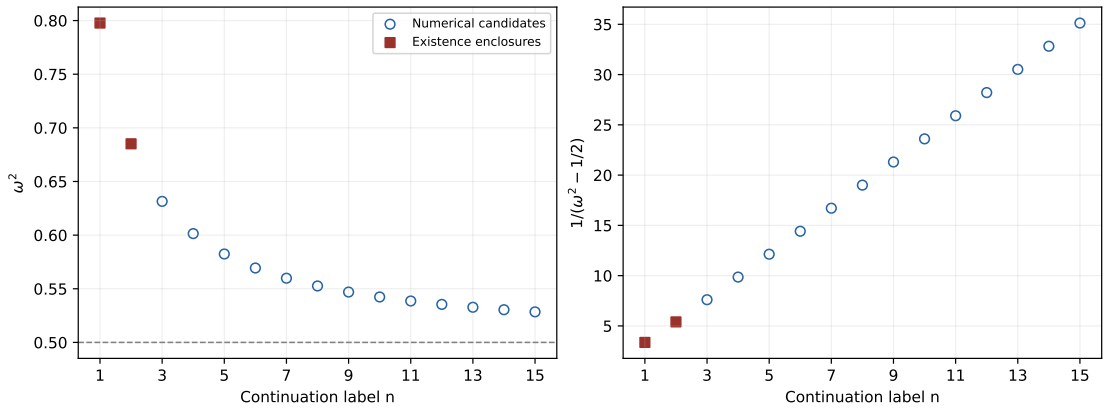


Figure 2: Archived radial locations through $n = 15$ and their inverse detunings $1/(\omega^2 - 1/2)$. Filled symbols mark C0/C1. The plot retains its original fifteen-point dataset, including the earlier rounded index-15 estimate; Table 2 records the subsequent 15–18 calculation explicitly. The supplementary source inventory binds the figure to its data. No new fit or extrapolated points are added.

September 2026, 08:33:26 CEST, and its SHA256 binding are included in the supplementary source inventory. This is a documented calibrated extrapolation, not global preregistration. The five later comparisons passed their respective diagnostic windows, but showed a systematic thin-wall offset of about 2×10^{-5}, comparable to those windows. The unfitted planar comparison below therefore carries the mechanistic interpretation; the forecast history does not establish zero linewidth.

5 Thin-wall phase matching and a planar transmission zero

Put $\epsilon = \omega^2 - \omega_c^2$ and $S_c = 1/(2\beta)$. The leading thin-wall profile and radius in three dimensions are

$$R_{\mathrm{tw}} \simeq \frac{1}{2\sqrt{\beta}\,\epsilon}, \qquad f^2(r) \simeq \frac{S_c}{1 + \exp[(r - R_{\mathrm{tw}})/\sqrt{\beta}]}. \tag{15}$$

These are asymptotic approximations, not substitutes for the certified profiles. The leading radius relation is established prior work: with $\Delta = m_\phi^2 - \omega_0^2$, Heeck et al. use $\kappa^2 = (\omega^2 - \omega_0^2)/\Delta$ and $R^* = \sqrt{\Delta} R$, and obtain $R^* \simeq 1/\kappa^2$ in their Eq. (40) [22]. Here $\Delta = 1/(4\beta)$, which gives the radius in Eq. (15). Their inflection-point radius and our half-density radius need not have identical subleading coefficients. Their Eq. (58) gives $R^* = \kappa^{-2} + \frac{1}{4} - \frac{5}{16}\kappa^2 + O(\kappa^4)$; these terms require a radius-convention conversion before comparison with the present half-density radius [22]. Diagonalizing the constant interior channel matrix gives the propagating interior wavenumber

$$k_{\mathrm{in}}^2 = \omega^2 + \rho^2 - D_c + \sqrt{4\omega^2\rho^2 + C_c^2}, \qquad D_c = 1 + \frac{1}{4\beta}, \quad C_c = \frac{1}{2\beta}. \tag{16}$$

At the stated interior amplitude, this is the same dispersion branch as Eq. (101) of Zhang, Zhou, and Zhu [8]; the interior wavenumber is an established ingredient, not a new result here. A regular interior standing wave reaches the wall with phase $\Psi = k_{\mathrm{in}} R_{\mathrm{tw}}$. The bare closed-channel wall approximation supplies a localized state, while matching it to the open interior wave permits a sign-changing radiation amplitude. A useful leading parametrization is

$$k_{\mathrm{in}}(\omega_n, \rho_n) R_{\mathrm{tw}}(\omega_n) \simeq \pi[n + \theta(\epsilon_n)], \qquad \rho_n \simeq \omega_n + c(\epsilon_n). \tag{17}$$

The radius and interior dispersion follow from the model approximations; the wall phase θ and dressed level shift contained in c remain calibrated. A separate coupled planar-wall calculation now supplies a numerical transmission zero, but does not determine both finite-curvature functions

independently. Consequently, agreement of the calibrated phase rule with the sequence is not a parameter-free prediction.

For $\beta = 1/2$ and $\omega^2 = 1/2$, the exact planar profile is $S(x) = [1 + \exp(\sqrt{2}x)]^{-1}$. Solving the two-channel scattering problem, with one open and one evanescent channel on each side, gives

$$\rho_z \simeq 1.52414976213, \qquad k_{\mathrm{in}}(\rho_z) \simeq 1.92332459129. \tag{18}$$

An independently implemented Numerov boundary-value calculation confirmed this zero before reading the comparison calculation. It used a continuously phase-normalized transmission amplitude, actual complex-transmission residuals, flux and reciprocity controls, and separate mesh and boundary refinements. A conservative empirical frequency uncertainty is 10^{-9}; this is not a certified enclosure. Two finite frequency scans found no additional zero, but do not prove uniqueness throughout the open-channel window. The planar scattering state has an incident and reflected wave; it is not itself a square-integrable embedded mode.

If $k_{\mathrm{in}} \to k_\infty > 0$ and the wall phase has a finite limit, Equation (17) predicts, within the approximation,

$$\epsilon_n \sim \frac{1}{b_\infty n}, \qquad b_\infty = \frac{2\sqrt{\beta}\,\pi}{k_\infty}. \tag{19}$$

From the historical rounded locations 0.535449, 0.532772, 0.530470, 0.528469, the inverse-detuning steps for indices 12–15 are approximately 2.3043, 2.3053 and 2.3068; these descriptive values do not specify accuracy beyond the input locations. The calibrated model gives 2.298, while its bare wall-state estimate gives 2.334. Substituting the independently computed planar zero into Equation (19) instead gives $b_\infty^{\mathrm{wall}} \simeq 2.31000162857$, with a conservative empirical numerical uncertainty of 2×10^{-9}. This wall-based candidate was not fitted to ladder locations. Identifying it with the actual radial limiting step still requires a controlled curvature and phase-matching limit. This supports a useful spacing description; it does not prove existence of infinitely many localized modes or identify an exact limiting constant. The phase model also motivates alternating signs of a matching Jacobian and quadratic linewidth minima, but neither property independently replaces a tail-controlled existence proof. The inverse-first-power scaling follows from the calibrated model, not from an independently established asymptotic exponent. Exploratory free power fits yielded exponents between -0.76 and -0.88; the reciprocal fit improved the comparison metric only by factors of approximately 1.2–2, below the originally sought factor of ten. Finite-data agreement therefore does not establish an exact asymptotic law.

A local numerical check also supports a planar transmission cancellation for the modified potential $U(S) = S - S^2 + S^3$ at $\omega_{\mathrm{min}} = \sqrt{3}/2$. An independent Numerov implementation, with the candidate frequency already known, gives $\rho_z \simeq 1.77345307180$ and $k_{\mathrm{in}} \simeq 2.3994321234$. The last grid change, $h = 0.04$ to 0.02, shifts ρ_z by 1.64×10^{-10}, while increasing the half-domain from 32 to 40 at the finest spacing changes it by 4.2×10^{-14}; these are empirical sensitivities, not certified bounds. The corresponding thin-wall spacing candidate is $2\pi/k_{\mathrm{in}} \simeq 2.618613482$. This known-target local replication neither counts all planar zeros nor establishes the radial ladder limit for the modified potential.

The corresponding radial calculation is also available. Its last previously reported inverse-detuning step was 2.596420, about 0.9% below the planar candidate. A subsequent search gave the following finite-domain numerical locations; k continues that study's local index and is not a certified node count:

k	ω^2	$1/(\omega^2 - 3/4)$	Resolved winding
0	0.778585440	34.98284	-1
-1	0.776606628	37.58462	$+1$
-2	0.774882776	40.18844	-1

The reported empirical uncertainties are about 2×10^{-8} in ω^2 and 3×10^{-5} in inverse detuning. These later increments are approximately 2.6018 and 2.6038; convergence to the planar value

remains a conjectural interpretation. A frozen absolute-location formula truncated after the $O(1)$ term failed its combined test: inverse-detuning errors for this potential were 0.23–0.26, outside the preassigned 0.10 window; an extrapolation of the preceding radial locations was more accurate for both potentials. This failure concerns the finite-order absolute-location formula, not a proof or disproof of the leading wall spacing.

6 Nonlinear sources and the limits of linear trapping

A localized eigenmode of the linearized equation need not remain radiationless at finite amplitude. We distinguish an exact source calculation, the kinematic opening of radiation channels, and a nonzero outgoing response. Existing nonlinear evidence for radial location C0 is described below; it is not transferred to the additional modes C1 and C2 without evaluating their own sources.

6.1 Quadratic expansion

We use the complex nonlinear Klein–Gordon equation

$$\phi_{tt} - \Delta\phi + U'(|\phi|^2)\phi = 0, \qquad U(S) = S - S^2 + \tfrac{1}{2}S^3, \qquad \phi = e^{i\omega t}(f + z), \tag{20}$$

where the real background satisfies $-\Delta f - \omega^2 f + U'(f^2)f = 0$. Setting $s = f^2$, the linear operator is

$$\mathcal{L}z = z_{tt} + 2i\omega z_t - \Delta z + [U'(s) + sU''(s) - \omega^2]z + sU''(s)\overline{z}. \tag{21}$$

The quadratic term in the equation's left-hand side is exactly

$$\begin{aligned} G_2[z] &= fU''(s)(z^2 + 2|z|^2) + \tfrac{1}{2}f^3U'''(s)(z + \overline{z})^2 \\ &= \mathcal{A}z^2 + 2\mathcal{A}|z|^2 + \mathcal{C}\overline{z}^{\,2}, \qquad \mathcal{A} = f\left(-2 + \tfrac{9}{2}f^2\right), \qquad \mathcal{C} = \tfrac{3}{2}f^3. \end{aligned} \tag{22}$$

Consequently $z = \epsilon z_1 + \epsilon^2 z_2 + \cdots$ gives $\mathcal{L}z_1 = 0$ and $\mathcal{L}z_2 = -G_2[z_1]$. In particular, the forcing has the opposite sign to the Taylor term in Eq. (22).

For a real spherical harmonic Y and real radial functions a, b, write

$$z_1 = Y\left(ae^{i\rho t} + be^{-i\rho t}\right). \tag{23}$$

Then

$$G_2[z_1] = Y^2\left(Q_+e^{2i\rho t} + Q_0 + Q_-e^{-2i\rho t}\right), \tag{24}$$

$$Q_+ = \mathcal{A}(a^2 + 2ab) + \mathcal{C}b^2, \qquad Q_- = \mathcal{A}(b^2 + 2ab) + \mathcal{C}a^2, \tag{25}$$

$$Q_0 = 2\mathcal{A}(a^2 + b^2) + 2(\mathcal{A} + \mathcal{C})ab. \tag{26}$$

These expressions introduce rotating-frame frequencies $0, \pm 2\rho$, or laboratory frequencies $\omega, \omega \pm 2\rho$. The zero-frequency term is static only in the rotating frame.

6.2 Angular selection and forced channel problems

For C1, the radial $l = 0$ mode, $Y = Y_{00}$ and $Y_{00}^2 = Y_{00}/\sqrt{4\pi}$, so only $L = 0$ occurs at this order. For C2, choose a real normalized $l = 1$ harmonic. After a spatial rotation it equals Y_{10}, and

$$Y_{10}^2 = \frac{Y_{00}}{\sqrt{4\pi}} + \frac{Y_{20}}{\sqrt{5\pi}}. \tag{27}$$

Thus each of the three temporal source coefficients can drive $L = 0$ and $L = 2$; the antisymmetric $L = 1$ product is absent. A different real orientation rotates the quadrupole within the $L = 2$ multiplet. This standing real mode must not be confused with a rotating complex $Y_{11}/\overline{Y}_{11}$ block: the latter has $Y_{11}^2 = \sqrt{3/(10\pi)}Y_{22}$ for its corresponding oscillatory product.

Let c_L denote the coefficient in the relevant angular product and $D_L = -\partial_r^2 - 2r^{-1}\partial_r + L(L+1)r^{-2}$. For the oscillatory response $h_{+,L}e^{2i\rho t} + h_{-,L}e^{-2i\rho t}$ one must solve the coupled system

$$\begin{pmatrix} D_L + U' + sU'' - (\omega+2\rho)^2 & sU'' \\ sU'' & D_L + U' + sU'' - (\omega-2\rho)^2 \end{pmatrix} \begin{pmatrix} h_{+,L} \\ \overline{h}_{-,L} \end{pmatrix} = -c_L \begin{pmatrix} Q_+ \\ Q_- \end{pmatrix}. \tag{28}$$

Here U', U'' are evaluated at s; the conjugated second unknown matters when imposing outgoing conditions. The real static correction obeys $[D_L + U' + 2sU'' - \omega^2]h_{0,L} = -c_L Q_0$. Gauge and charge constraints must also be fixed in constructing a nonlinear continuation.

Both C1 and C2 have $|\omega| < 1$ and $|\omega \pm 2\rho| > 1$. For C2, the latter frequencies are approximately 4.52130 and -2.78407. The second harmonics are therefore open in the mass-one vacuum, whereas the rotating-static correction is closed. The centrifugal term does not change the asymptotic mass threshold. With the convention e^{iEt}, an outgoing wave has radial phase $e^{-i\,\mathrm{sgn}(E)\sqrt{E^2-1}\,r}$; a negative E does not mean negative outward energy transport.

An open channel and a nonzero local source do not establish radiation: the outgoing source overlap can vanish. A localized continuation without incoming radiation requires both open amplitudes to vanish in every allowed L block of Eq. (28). The C1 response remains open; an approximate-input C2 calculation is reported below. A nonzero such amplitude would obstruct a regular localized perturbative continuation with the specified leading mode, but would not exclude all other nonlinear states. Frequency shifts of order ϵ^2 do not automatically cancel the second harmonic: the carrier shift contributes a static background correction at this order, while a shift of ρ first acts on ϵz_1 at order ϵ^3.

6.3 Complex dipole polarization and a conditional obstruction

The threefold dipole degeneracy allows more than rotations of a real standing mode. Let $Y_i(\boldsymbol{n}) = \sqrt{3/(4\pi)}n_i$ be real orthonormal Cartesian dipole harmonics and write $A(\boldsymbol{n}) = \sum_i p_i Y_i(\boldsymbol{n})$, with $\boldsymbol{p} \in \mathbb{C}^3$. For the same real radial eigenmode as above, the allowed tangent has the linked sidebands $z_1 = a(r)Ae^{i\rho t} + b(r)\overline{A}e^{-i\rho t}$. This does not allow independently chosen polarizations of the two sidebands or a mixture of distinct radial modes.

Set $\nu = \boldsymbol{p}^\dagger\boldsymbol{p}$ and $\chi = \boldsymbol{p}^T\boldsymbol{p}$. The oscillatory quadratic forcing is proportional to A^2, rather than $|A|^2$. Its orthogonal angular projections satisfy the exact identities

$$\|P_0(A^2)\|^2_{L^2(S^2)} = \frac{|\chi|^2}{4\pi}, \qquad \|P_2(A^2)\|^2_{L^2(S^2)} = \frac{3\nu^2 - |\chi|^2}{10\pi} \geq \frac{\nu^2}{5\pi}. \tag{29}$$

These follow by decomposing $\boldsymbol{p}\boldsymbol{p}^T$ into its trace and symmetric traceless parts and integrating the fourth spherical moment. Circular polarization, for example $\boldsymbol{p} = (1, i, 0)/\sqrt{2}$, has $\chi = 0$ and hence no oscillatory $L = 0$ source, but its $L = 2$ source does not vanish.

This source identity has a conditional implication for the outgoing response. Define $C_{\mathrm{ref}} \in \mathbb{C}^2$ to be the *exact* radial $L = 2, m = 0$ outgoing coefficient for the normalized Y_{10} tangent, with the same radial normalization. Assuming the exact regular outgoing problem exists uniquely, spherical symmetry makes its radial operator independent of m. If $b_m = \langle Y_{2m}, A^2\rangle$, linearity gives

$$C_{2m} = \sqrt{5\pi}\, b_m C_{\mathrm{ref}}, \qquad \sum_m \|C_{2m}\|^2 = \frac{3\nu^2 - |\chi|^2}{2}\|C_{\mathrm{ref}}\|^2 \geq \nu^2\|C_{\mathrm{ref}}\|^2. \tag{30}$$

The same identity holds separately for either radial component. Conjugating the second computational channel to the physical sideband permutes conjugate spherical harmonics and preserves these norm sums. Thus an exact radial $L = 2$ nonzero certificate would cover every nonzero complex polarization in this one radial dipole eigenspace. The numerical coefficients reported below do not provide that certificate.

For clarity, the corresponding nonlinear obstruction requires a precise continuation class. Suppose a family is twice continuously differentiable in amplitude as a 2π-periodic rotating-frame

profile in $C^2(\mathbb{T}; H^4_\gamma(\mathbb{R}^3))$, where $H^4_\gamma = \{w : e^{\gamma\sqrt{1+|x|^2}} w \in H^4\}$. For the exact C2 background and tangent we may fix $\gamma = 1/10$: the inherited exterior decay rates are at least 0.495 and 0.287, respectively. Their radial equations propagate this decay to derivatives through order four; the regular origin expansions are even for the background and of the form r^2 times even functions for the reduced dipole components. Consequently the unreduced Cartesian fields are smooth at the origin and belong to this same weighted space. This membership statement does not assume that a nonlinear continuation exists or certify a numerical Sobolev norm. Allow smooth carrier and modulation frequency shifts and an additional static first tangent, but no other first-order temporal harmonics. A nonzero exact second-harmonic outgoing amplitude would contradict localization of the second amplitude derivative. At this order the frequency-shift terms are static or fundamental, so they cannot remove the second-harmonic source. This excludes only that uniformly localized smooth continuation class; it is not a dynamical instability or lifetime theorem.

6.4 Numerical response and exact-mode comparison

Approximate-input forced solves in the monopole and quadrupole sectors found nonzero second-harmonic outgoing coefficients. The power table, input-sensitivity comparisons and source–current consistency check are retained in Supplement Section S2.1. These use a numerical unit-L^2 normalization. The exact-mode transfer below instead uses Jost normalization and bounds all terms of the coefficient-error identity, including the infinite exterior.

6.5 Monopole coefficient enclosure and its dependencies

For the real dipole tangent at C2, we assembled a complete error budget for the $L = 0$ second-harmonic forced problem. Here L labels the radiated angular sector, whereas the underlying localized linear mode has $l = 1$. The calculation uses the exact-mode Jost normalization, with the evanescent reduced component satisfying $\lim_{r\to\infty} e^{\kappa_c r} v(r) = 1$. It therefore has a different amplitude normalization from the numerical power table in Section S2.1.

The error identity includes the inner and outer source differences, operator differences, the residual of the fixed approximate response, its actual matching jumps and origin contribution, and the exterior boundary and coefficient-extraction errors. The exterior potential and source are included through infinity. No measured small linewidth or comparison between two numerical resolutions substitutes for these terms.

The following inherited hypotheses specify the exact problem to which the numerical error budget applies; they are not new conclusions of the scalar summation:

1. The C2 existence enclosure of Section 3 supplies one common exact parameter tuple $z_* = (a_*, c_*, \rho_*, \omega_*)$ in its certified dyadic box, a positive regular spherical profile f_*, and a nonzero real regular dipole mode. All profile, mode and parameter-error bounds refer to this same tuple. Neither uniqueness of the tuple nor the claim that every point of the box is an exact root is assumed.

2. At that exact tuple, the decaying fundamental Jost solution belongs to the span of the two regular origin solutions. This is the inherited regular/Jost identification from the C2 Wronskian conditions, not an inference from overlap of numerical residual intervals. Its reduced components are $O(r^2)$ at the origin and obey the same normalization $e^{\kappa_c r} v(r) \to 1$, with $\kappa_c = \sqrt{1 - (\omega_* - \rho_*)^2}$. The background and mode satisfy the analytic infinite-tail bounds: for $r \geq 36$ one may use $|f_*(r)| \leq (97/8)e^{-0.495r}/r$ and $|u(r)| + |v(r)| \leq 1.097e^{-0.287r}$.

3. The exact quadratic $L = 0$ source is the complete product source of the real standing dipole in Eqs. (22)–(28). The fixed nominal mode, source, response and comparison coefficient use the same amplitude convention: the nominal unit-L^2 mode is divided by the exact rational $s_0 = 2558188928468177/2251799813685248$, and quadratic quantities by s_0^2. This does not

identify s_0 with the norm of the exact mode. The computational carrier conjugation is retained throughout.

4. For the second harmonic, the exact reduced operator is $-\partial_r^2 - Q_*^2 + V_*(r)$ with real symmetric, locally bounded V_* and $Q_* = \mathrm{diag}(q_+, q_-)$, $q_\pm = \sqrt{(\omega_* \pm 2\rho_*)^2 - 1}$. Both channels satisfy $5/2 < q_\pm < 23/5$. The exterior Volterra bounds hold at these exact frequencies, with the analytic potential and source tails retained through infinity. At $R = 64$, the exterior potential satisfies $\int_R^\infty \|V_*(r)\|_2 \, dr/(5/2) < 1/2$ and the source is integrable. Thus the exact outgoing reduction includes both its homogeneous maps and its inhomogeneous source corrections. Together with regular Dirichlet data at the origin, the positive outgoing matrix current gives the unique forced solution and adjoint extraction problem. This argument concerns two open second-harmonic channels, not the fundamental embedded mode.

5. The analytic and validated input bounds used in the error identity enclose the entire core and exterior, including exact-to-nominal frequency and map changes. In Euclidean vector and induced matrix norms, the inherited bounds $\int_0^\infty \|V_*(r)\|_2 \, dr < 25/4$ and the Volterra estimates give the adjoint amplitude weights $\|Z\|_2 < 26/5$ and $\|Z'\|_2 < 14$. Source and operator differences, response residuals, origin and matching jumps, exterior source terms, and coefficient-extraction rounding are all bounded in this same convention, without dropping a tail or counting one twice. The directed-arithmetic implementation and its interval libraries remain part of the computational trust assumptions.

Supplement Section S1 assigns each item to its analytical lemma and numerical record, distinguishing established dependencies from software trust and the assumed continuation class. The qualification is about that inherited chain; it does not denote an omitted term in the completed coefficient-error budget. Under these inherited hypotheses, the computed common coefficient-error bound is

$$\|C_* - C_{\mathrm{num}}\|_2 \le \mathcal{E} < 0.001023269269845280. \tag{31}$$

The frozen comparison coefficient in the second computational channel has the lower bound

$$|C_{\mathrm{num},2}| > 0.001582591541472706. \tag{32}$$

The directed arithmetic establishes, before rounding for display,

$$|C_{*,2}| > 0.000559322271627427. \tag{33}$$

This channel has laboratory frequency $\omega_* - 2\rho_* < 0$. The solver uses a negative carrier and conjugates its second computational component to recover that physical sideband. Conjugating the entire field to the positive-carrier convention used here gives $C_{-,\mathrm{paper}} = C_{*,2}$; all these operations preserve the modulus. The first channel remains undecided by this common error bound; that failure does not imply its coefficient vanishes.

This result supplies the nonzero-coefficient premise of the conditional smooth-continuation argument for every linked dipole polarization with $\boldsymbol{p}^T\boldsymbol{p} \neq 0$. The complex-polarization coefficient is proportional to this quantity and therefore does not cover circular polarization. It gives neither a nonlinear lifetime nor a dynamical instability theorem. In particular, the bound is not a certified value for any power coefficient in the supplementary approximate-input table.

6.6 Earlier radial evidence

At C0, finite-amplitude radial evolutions support fourth-order scaling of a nonlinear-remainder flux and an outgoing second harmonic. This power counting is familiar from internal-mode radiation, for example in the one-dimensional kink calculation of Manton and Merabet [36, Appendix A]; the present calculations test it in a different model and spatial dimension. A separate forced

frequency-domain calculation agrees with the dominant measured harmonic within its prescribed criteria. A frozen finite-operator interval calculation separates one necessary overlap from zero, while its original two-overlap acceptance test failed. Its spline inputs and free exterior are fixed data, so it does not certify that overlap for the exact continuum mode. Supplement Section S2.2 retains these numerical values and limitations.

7 Discussion and conclusion

The strongest result is the existence of three localized linear modes with analytic treatment of the infinite exterior. The remaining radial locations have the numerical evidence classes recorded in Table 2. The unfitted planar spacing candidate provides a mechanism to test; a controlled radial curvature limit and an infinite-ladder theorem remain open. The failed absolute-location test in Section 5 constrains the finite-order approximation.

The conditional monopole enclosure obstructs uniformly localized smooth continuation for the linked dipole polarizations with $\boldsymbol{p}^T\boldsymbol{p} \neq 0$. A complete quadrupole certificate is still required for circular polarization. The numerical power table has a different normalization and is not recertified by this bound. None of these statements supplies a nonlinear lifetime or a stability theorem for the background. In particular, the charge slope $dQ/d\omega$ at the three certified backgrounds has not been evaluated in this revision; a stability assignment from that criterion is left open. An independent interval implementation and an operator realization of the full evolution generator would strengthen the computational and spectral interpretation respectively.

The Supplement retains prescribed driving, formation diagnostics, discretization sensitivity and the numerical response controls as exploratory material. They supply no additional existence enclosures.

Data and code availability

The supplementary archive, prepared from the preserved historical records, contains proof code, certificates, selected numerical records, source bindings and environment instructions. It is distributed as the file `qball-bic-v0.51-archive-r2.zip` of release `qball-bic-v0.51` at https://github.com/fmhc/physics/releases/tag/qball-bic-v0.51. This is a selected-records archive, not a complete reproducibility package for every numerical statement. The source and matching records for ladder indices 3–5, the separate $\beta = 1$ ladder and wall studies, and the data package assembling the linewidth reconstructions are not included. The nonlinear and exploratory programs are historical source records rather than portable rerun packages; parent computations of the exploratory results are not packaged. The archive README's coverage table specifies these limits. Historical checksums and the checksums of metadata-redacted distribution copies are recorded separately; such redaction does not alter numerical fields or replace the historical provenance. Text, figures, data and certificates are licensed under CC BY 4.0; program code is licensed under the MIT License. For this revision, the three primary linear certificate runs were repeated with unchanged proof code. After excluding four timing fields, their receipts agree exactly with the archived receipts; all point and box step records are byte-identical. This is reproduction of the same implementation, not an independent interval implementation. The remaining historical numerical calculations have not been repeated for this revision. The proof appendix and the supplementary hypothesis-by-hypothesis audit specify the analytical dependencies.

Use of AI tools

This work was carried out with extensive use of AI tools: Claude (Anthropic) and Codex (OpenAI). They were used for drafting text, writing and running code, the interval computations, literature comparison and internal reviews. The author chose the questions, directed the work, decided on

scope and publication, and takes responsibility for the content. In the companion archive, names of individual AI agents in historical records are replaced by neutral labels. The manuscript has not been peer reviewed.

A Analytic lemmas underlying the linear certificates

This section spells out the analytical implications tested by the certificates. It concerns three spatial dimensions and radial angular sectors $\ell = 0, 1$ only. A radial reduction of the three-dimensional problem is not a one-dimensional field theory. The normalization, origin powers and exterior kernels below depend on dimension and angular momentum; they cannot be transferred unchanged to other cases. All inequalities are evaluated over the whole dyadic parameter box.

A.1 Profile tail: contraction and matching errors

Let $1/\sqrt{2} < \omega < 1$, $\kappa_0 = \sqrt{1-\omega^2}$, $L, c, \bar{f} > 0$, and define

$$K_P = \frac{(2 + \frac{3}{2}\bar{f}^2)e^{-2\kappa_0 L}}{4\kappa_0^2 L^2}, \qquad \eta = 2K_P c^3, \qquad m = c + \eta,$$
$$\Lambda = \frac{(6 + \frac{15}{2}\bar{f}^2)m^2 e^{-2\kappa_0 L}}{4\kappa_0^2 L^2}. \tag{34}$$

Assume $me^{-\kappa_0 L}/L \le \bar{f}$, $K_P m^3 \le \eta$, $\Lambda < 1$ and $c > \eta$. Then the profile Volterra equation has a unique solution in $\|\widetilde{u} - c\|_\infty \le \eta$, where $f = e^{-\kappa_0 r}\widetilde{u}/r$. It is positive and solves the profile equation. With $\eta_0 = K_P m^3$, its matching errors satisfy

$$|Le^{\kappa_0 L} f(L) - c| \le \eta_0, \qquad |e^{\kappa_0 L}(Lf'(L) + f(L)) + \kappa_0 c| \le 2\kappa_0 \eta_0. \tag{35}$$

Proof. On the indicated complete ball in $C_b([L,\infty))$, $|\widetilde{u}| \le m$ and $|f| \le \bar{f}$. The kernel is bounded by $1/(2\kappa_0)$ and

$$\int_r^\infty e^{-2\kappa_0 s} s^{-2}\,\mathrm{d}s \le \frac{e^{-2\kappa_0 r}}{2\kappa_0 r^2}. \tag{36}$$

Thus the integral map differs from c by at most $K_P m^3$. The nonlinear integrand as a function of $\widetilde{u}$ is $-2e^{-2\kappa_0 s}s^{-2}\widetilde{u}^3 + \frac{3}{2}e^{-4\kappa_0 s}s^{-4}\widetilde{u}^5$; its derivative is bounded in modulus by $(6+\frac{15}{2}\bar{f}^2)m^2 e^{-2\kappa_0 s}s^{-2}$. The map is therefore a contraction of constant Λ. Banach's theorem supplies its fixed point and $\widetilde{u} \ge c - \eta > 0$. Writing $u = e^{-\kappa_0 r}\widetilde{u}$ gives

$$u(r) = ce^{-\kappa_0 r} + \int_r^\infty \frac{\sinh(\kappa_0(s-r))}{\kappa_0} q_f(s)u(s)\,\mathrm{d}s, \qquad q_f = -2f^2 + \tfrac{3}{2}f^4. \tag{37}$$

Differentiation proves $u'' - \kappa_0^2 u = q_f u$ and the profile equation. The first bound in (35) follows from the self-map estimate; for the second use

$$e^{\kappa_0 r}u'(r) + \kappa_0 c = -\int_r^\infty \frac{1 + e^{-2\kappa_0(s-r)}}{2} q_f(s)\widetilde{u}(s)\,\mathrm{d}s. \tag{38}$$

The same integral majorant gives $2\kappa_0\eta_0$ at L. The integral tails vanish at infinity, proving the asserted normalization. For a fixed profile-row scale s_1 and $E_0 = e^{-\kappa_0 L}$, the resulting matching-map corrections satisfy

$$|E_1| \le |s_1|E_0\eta_0/L, \qquad |E_2| \le |s_1|E_0(2\kappa_0\eta_0/L + \eta_0/L^2). \tag{39}$$

A.2 Decaying mode: the two angular sectors

Put $dV=-4f^2+\frac{9}{2}f^4$, $C=-2f^2+3f^4$ and $q_J=|dV|+|C|\le 6f^2+\frac{15}{2}f^4$. A common integral bound is

$$Q=\frac{(6+\frac{15}{2}\bar f^2)m^2e^{-2\kappa_0 L}}{2\kappa_0 L^2}\ge\int_L^\infty q_J(s)\,\mathrm{d}s. \tag{40}$$

Assume $k=\sqrt{(\omega+\rho)^2-1}>0$ and $\kappa_c=\sqrt{1-(\omega-\rho)^2}>0$. Normalize $(\widetilde A,\widetilde B)=e^{\kappa_c r}(A,B)$. For $\ell=0$ their Volterra equation has free term $(0,1)$ and kernels

$$K_A=e^{-\kappa_c(s-r)}\frac{\sin(k(s-r))}{k},\qquad K_B=\frac{1-e^{-2\kappa_c(s-r)}}{2\kappa_c}. \tag{41}$$

The first kernel multiplies $dV\widetilde A+C\widetilde B$ and the second multiplies $dV\widetilde B+C\widetilde A$. With $\mu=\max(1/k,1/(2\kappa_c))$ and $e_0=e^{\mu Q}-1$, one obtains

$$\begin{aligned}|\widetilde A|&\le\frac{e_0}{\mu k},&\qquad|\widetilde B-1|&\le\frac{e_0}{2\mu\kappa_c},\\ |e^{\kappa_c r}A'|&\le\left(\sqrt{1+\kappa_c^2/k^2}+\kappa_c/k\right)\frac{e_0}{\mu},&\qquad|e^{\kappa_c r}B'+\kappa_c|&\le\frac{3e_0}{2\mu}.\end{aligned} \tag{42}$$

These bounds hold for all $r\ge L$.

For $\ell=1$ define the free Riccati functions

$$\begin{aligned}j(x)&=\sin(x)/x-\cos(x),&\qquad y(x)&=-\cos(x)/x-\sin(x),\\ d(u)&=e^{-u}(1+1/u),&\qquad g(u)&=e^{u}(1-1/u),\\ G_A(r,s)&=\frac{j(kr)y(ks)-y(kr)j(ks)}{k},&\qquad G_B(r,s)&=\frac{d(\kappa_c r)g(\kappa_c s)-g(\kappa_c r)d(\kappa_c s)}{2\kappa_c}.\end{aligned} \tag{43}$$

Here $K_{A,B}=e^{-\kappa_c(s-r)}G_{A,B}$ and the free term is $(0,1+1/(\kappa_c r))$. For $kL,\kappa_cL\ge1$ put

$$\begin{aligned}a_1&=\frac{1+(kL)^{-2}}{k},&\qquad a_2&=\sqrt{1+(kL)^{-2}},\\ b_0&=1+(\kappa_cL)^{-1},&\qquad b_1&=\frac{b_0}{2\kappa_c},&\qquad b_2&=1+(\kappa_cL)^{-1}+\tfrac12(\kappa_cL)^{-2}.\end{aligned} \tag{44}$$

Then $|K_A|\le a_1$, $|K_B|\le b_1$, $|e^{-\kappa_c(s-r)}\partial_r G_A|\le a_2$ and $|(\partial_r-\kappa_c)K_B|\le b_2$. With $\mu=\max(a_1,b_1)$ and $e_1=b_0(e^{\mu Q}-1)/\mu$ the four errors in normalized physical Cauchy data are bounded by $(a_1,b_1,a_2,b_2)e_1$ about

$$J_0=\left(0,1+\frac{1}{\kappa_cL},0,-\kappa_c-\frac{1}{L}-\frac{1}{\kappa_cL^2}\right). \tag{45}$$

In particular $e^{\kappa_c r}(|A|+|B|)\le b_0+(a_1+b_1)e_1$.

Proof. For $\ell=0$, the kernels have bounds $1/k$ and $1/(2\kappa_c)$. The nth ordered Volterra iterate is bounded by $(\mu Q)^n/n!$. Thus the series converges uniformly and $\psi=\max(|\widetilde A|,|\widetilde B|)\le e^{\mu Q(r)}$, where $Q(r)=\int_r^\infty q_J$. Since $\int_r^\infty q_J(s)e^{\mu Q(s)}\,\mathrm{d}s=(e^{\mu Q(r)}-1)/\mu$, the value bounds follow. Differentiating the kernels gives $|\partial_r K_A|\le\sqrt{1+\kappa_c^2/k^2}$ and $|\partial_r K_B|\le1$; subtracting κ_c times the normalized values gives the displayed conservative physical derivative bounds. Twice differentiating the integral equations proves the radial ODE.

For $\ell=1$, direct differentiation gives $W_x(j,y)=1$, $j^2+y^2=1+x^{-2}$ and $j'^2+y'^2=1-x^{-2}+x^{-4}\le1$ for $x\ge1$. Cauchy–Schwarz yields a_1,a_2. The normalized closed kernel is

$$K_B=\frac{(1+1/u)(1-1/v)-e^{-2(v-u)}(1-1/u)(1+1/v)}{2\kappa_c},\quad v=\kappa_cs\ge u=\kappa_cr\ge1. \tag{46}$$

Both terms are nonnegative before subtraction; the first dominates, so $0 \le K_B \le b_0/(2\kappa_c)$. Differentiating gives the stated b_2 bound (it concerns $(\partial_r - \kappa_c)K_B$, not $\partial_r K_B$). More explicitly, twice the absolute normalized physical derivative kernel is

$$(1+u^{-1}+u^{-2})(1-v^{-1}) + e^{-2(v-u)}(1-u^{-1}+u^{-2})(1+v^{-1}) \le 2 + 2u^{-1} + u^{-2}. \tag{47}$$

The inequality follows by replacing the exponential by one, leaving $2 + 2u^{-2} - 2/(uv)$, which is at most the right-hand side for $u \ge 1$. The ordered-series argument now starts with a free term bounded by b_0. It gives $\psi \le b_0 e^{\mu Q(r)}$ and the four errors above. In both sectors the correction tends to zero, so $e^{\kappa_c r} B \to 1$ and the Jost solution is nonzero. If $(E_A, E_B, E_{A'}, E_{B'})$ denotes these four normalized Cauchy-data error bounds and s_3 is the fixed Wronski-row scale, then

$$|E_{2+i}| \le |s_3|(E_A|A_i'(L)| + E_B|B_i'(L)| + E_{A'}|A_i(L)| + E_{B'}|B_i(L)|), \quad i = 1, 2. \tag{48}$$

The core values on the right are enclosed over the entire parameter box. The exact dyadic scales s_1, s_3 change conditioning, not the zero set.

A.3 Continuity on the parameter box

The profile contractions use common upper bounds over the whole box, with $\inf \kappa_0 > 0$ and $\sup \Lambda < 1$. Their kernels vary continuously and have a common integrable majorant. For two parameter values z, z', the fixed points obey

$$\|\widetilde{u}_z - \widetilde{u}_{z'}\|_\infty \le \frac{1}{1-\Lambda} \sup_{u \in X'} \|(T_z - T_{z'})u\|_\infty \longrightarrow 0, \tag{49}$$

where a common ball X' contains both fixed points. Dominated convergence also gives continuity of the profile matching derivative. The Jost Picard iterates are continuous in parameters and converge uniformly with the common factorial bound. Their normalized Cauchy data are therefore continuous. Regular core solutions depend smoothly on parameters. Consequently every component of the tail correction E in the finite-dimensional matching problem is continuous.

A.4 Validated Taylor steps and the origin

Consider $y' = F(r, y)$ at a positive center r_j. Let a complex square $\mathcal{D}$ of half-side R enclose $|r - r_j| \le R$, and let B be a compact product of complex rectangles. Suppose F is analytic on a neighborhood of these sets and the interval test

$$Y_0 + \mathcal{D}_0 F(r_j + \mathcal{D}_0, B) \subset B, \qquad R < r_j, \tag{50}$$

is satisfied, where $\mathcal{D}_0 = \mathcal{D} - r_j$ and Y_0 encloses all initial values. If β_i bounds the distance of component B_i from its chosen center, the Taylor coefficients obey $|y_{i,n}| \le \beta_i R^{-n}$ for $n \ge 1$. For $N \ge 1$ and $0 < h < R$, $q = h/R$, the enclosed endpoint is

$$y_i(r_j + h) \in \sum_{n=0}^{N-1} y_{i,n} h^n + \left[-\frac{\beta_i q^N}{1-q}, \frac{\beta_i q^N}{1-q} \right]. \tag{51}$$

Proof. The Picard operator along straight complex segments maps analytic functions valued in B into the same set by (50). Convexity justifies the segment average. Analyticity supplies a finite Lipschitz constant on the enclosing set; on the disk $|r - r_j| \le R$, iterated differences have bound $(\Lambda_T R)^m/m!$. Picard iteration therefore converges uniformly to the unique analytic solution. Cauchy's coefficient bound for y_i minus the center of B_i, first on $R' < R$ and then as $R' \uparrow R$, gives the coefficient estimate. Summing the geometric series gives (51). Interval recursion encloses the retained coefficients. How the trial box was found does not affect this implication: only the

verified inclusion accepts the step. The condition $R < r_j$ keeps both $2/r$ and, for $\ell = 1$, $2/r^2$ analytic. It applies only to positive-center steps.

At the origin write $N(f) = (1-\omega^2)f - 2f^3 + \frac{3}{2}f^5$. The regular profile solves

$$f(r) = a + r^2 \int_0^1 \tau(1-\tau)N(f(\tau r))\,\mathrm{d}\tau, \qquad f'(r) = r\int_0^1 \tau^2 N(f(\tau r))\,\mathrm{d}\tau. \tag{52}$$

These formulas follow by integrating $(r^2 f')' = r^2 N(f)$ twice. They give the enclosure masses $R^2/6$ and $R/3$ and the recurrence $(n+2)(n+3)f_{n+2} = [N(f)]_n$. The derivative remainder uses its own box for f', not a differentiated endpoint error. Profile parameter variations obey the same integral operator with source $N'(f)\partial_p f + \partial_p N$. For the origin step let $\mathcal{D}_R$ enclose the complex disk $|r| \le R$, and let $\mathcal{D}_R^{[2]}$ enclose $\{r^2 : |r| \le R\}$. The validated a priori condition is

$$a + \tfrac{1}{6}\mathcal{D}_R^{[2]} N(B_f) \subset B_f, \qquad B_{f'} = \tfrac{1}{3}\mathcal{D}_R N(B_f). \tag{53}$$

It is checked componentwise over the full parameter box, with the same condition for each differentiated source. Only this verified self-inclusion permits the analytic Picard construction and the Cauchy bound for the start; the derivative remainder uses $B_{f'}$.

For $\ell = 0$, the mode equation is nonsingular at zero. For $\ell = 1$ set $A = r^2\alpha$, $B = r^2\beta$; the centrifugal term cancels and each regular component satisfies $y'' + 4y'/r = F$ with

$$y(r) = y(0) + \frac{r^2}{3}\int_0^1 \tau(1-\tau^3)F(\tau r)\,\mathrm{d}\tau, \qquad y'(r) = r\int_0^1 \tau^4 F(\tau r)\,\mathrm{d}\tau. \tag{54}$$

The enclosure masses are $R^2/10$ and $R/5$, with recurrence $(n+2)(n+5)y_{n+2} = F_n$, $y_1 = 0$. For the joint source box B, the corresponding verified componentwise conditions are

$$y(0) + \tfrac{1}{10}\mathcal{D}_R^{[2]} F(B) \subset B_y, \qquad B_{y'} = \tfrac{1}{5}\mathcal{D}_R F(B). \tag{55}$$

They apply also to the parameter-variation sources. Under these self-inclusions, Volterra iteration on the disk has factorial convergence as above; the Cauchy remainder applies to both y and y'. The two regular columns have initial data $(\alpha(0), \beta(0)) = (1,0), (0,1)$ and zero derivatives. At the first positive endpoint h_0, use the exact conversion $A = h_0^2\alpha$, $A' = 2h_0\alpha + h_0^2\alpha'$ (and likewise for B). The parameter-variation equations use the differentiated sources and the same origin construction. Thus the start is an enclosed regular solution, rather than prescribed approximate data at a small radius. For completeness, the core Jacobian calculation evolves the profile and its a, ω derivatives, the two regular mode columns, and their a, ρ, ω derivatives as one system of 38 real components. In particular, with $S = f^2$, $\partial_p S = 2f\partial_p f$, $\partial_p dV = (9S-4)\partial_p S$ and $\partial_p C = (6S-2)\partial_p S$. The explicit frequency derivatives are obtained from $-(\omega \pm \rho)^2$, while the profile source has $\partial_\omega N = -2\omega f$. The parameter c enters the profile tail matching, not the core initial-value problem.

A.5 Wronski matching, positivity and smoothness

For a symmetric radial matrix M, the bilinear form $W(X,Y) = X\cdot Y' - X'\cdot Y$ is constant because $W' = X\cdot MY - MX\cdot Y = 0$. For $\ell = 0$ choose $R_i(0) = 0$, $R_i'(0) = e_i$; then $W(\Psi, R_i) = \Psi_i(0)$. For $\ell = 1$, the Frobenius exponents are 2 and -1. Choose regular columns $R_i \sim r^2 e_i$ and singular columns $S_i \sim r^{-1}e_i/3$. Since $W(r^{-1}, r^2) = 3$, $W(S_i, R_j) = \delta_{ij}$ and $W(R_i, R_j) = 0$. Possible higher-order logarithmic terms do not change these limiting pairings. Therefore $W(\Psi, R_1) = W(\Psi, R_2) = 0$ removes the two singular coefficients. Profile matching equates the inner and outer Cauchy data; ODE uniqueness then gives one global profile and the Jost mode belongs to the regular origin span.

Suppose interval enclosures give $f > 0$ on $[0, L_p]$ and $|f| < \sqrt{S_-}$ on $[L_p, L]$, where

$$S_- = (2 - \sqrt{4 - 6\kappa_0^2})/3, \qquad 4 - 6\kappa_0^2 > 0. \tag{56}$$

Then $V_f = \kappa_0^2 - 2f^2 + \frac{3}{2}f^4 > 0$ on that remaining interval. With $u = rf$, $u'' = V_f u$ and both endpoint values positive, a negative interior minimum would have $u'' = V_f u < 0$, a contradiction. A zero minimum has $u = u' = 0$ and forces the identically zero solution, also a contradiction. The positive exterior fixed point completes $f > 0$.

The origin series for f is even. For $\ell = 0$, the regular mode components A, B are odd, so $A/r, B/r$ are even analytic functions. For $\ell = 1$, α, β are even analytic and $(A/r)Y_{1m} = \alpha(r)(rY_{1m})$ is smooth, since rY_{1m} is linear in Cartesian coordinates. The ODE and the exponentially decaying exterior bounds propagate decay to derivatives. This establishes smooth exponentially localized spatial fields in the two stated sectors.

A.6 Geometric radial simplicity and its extra inequality

Assume additionally $2\kappa_0 > \kappa_c$. Every solution Φ that is square integrable on $[L, \infty)$ is a multiple of the constructed Jost solution Ψ.

Proof. Bounded coefficients imply $\Phi'' \in L^2$. The interpolation estimate on overlapping unit intervals gives $\Phi' \in L^2$; hence $\Phi, \Phi' \to 0$. Two oscillatory comparison solutions can be constructed by Volterra iteration about $(\cos kr, 0), (\sin kr, 0)$ for $\ell = 0$, or $(j(kr), 0), (y(kr), 0)$ for $\ell = 1$. In the closed component the unnormalized Green kernel grows at most as a constant times $e^{\kappa_c(s-L)}$, while the coupling is bounded by a constant times $e^{-2\kappa_0 s}/s^2$. The assumption makes their product integrable. The ordered Picard series and its differentiated series converge, with corrections tending to zero. Their mutual Wronskian is $k \neq 0$; both are Wronski-orthogonal to the decaying Ψ. They and Ψ span a three-dimensional subspace of the four-dimensional space of radial Cauchy data: a nonzero combination of the oscillatory asymptotes cannot decay. Since $\Phi, \Phi' \to 0$, Φ is Wronski-orthogonal to this whole subspace. The Wronski form on Cauchy data is nondegenerate, so this orthogonal complement is one-dimensional and contains Ψ. It is precisely its span.

The inequality $2\kappa_0 > \kappa_c$ is needed for this comparison construction, not inferred from existence of the localized mode. For C0 the recorded lower bound on $2\kappa_0 - \kappa_c$ is 0.375236, separately checked but omitted from the historical aggregate success flag. The C1/C2 drivers include the condition. This appendix retains that audit qualification and does not silently replace the C0 driver. The conclusion is geometric simplicity in a fixed radial sector; the $\ell = 1$ angular degeneracy, algebraic multiplicity, Jordan chains and stability are separate questions.

A.7 Brouwer inclusion with a continuous correction

Let $Z = z_0 + [-\delta, \delta]$, $H = H_0 + E$ continuous, $H_0 \in C^1$, $DH_0(Z) \subset D$ and $|E| \leq \epsilon$ componentwise. The inclusion and weighted matrix bound in the main text imply a zero of H in Z. Indeed, the mean-value formula encloses $z - YH_0(z)$ by $z_0 - YH_0(z_0) + (I - YD)[-\delta, \delta]$; adding $-YE$ gives a self-map of Z. Brouwer supplies z_* with $YH(z_*) = 0$. Let $A = DH_0(z_0)$. The weighted row bound implies the operator norm of $I - YA$ is less than one in the box-weighted maximum norm. Thus YA is invertible, as is Y, and $H(z_*) = 0$. Neither differentiability of E nor uniqueness of the parameter zero follows from this argument.

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